



E/I/I test tables

Haul truck and light vehicle collision, Bindarra Gold Mine

Exported 4 October 2026

INVESTIGATION WORKFLOW SUITE

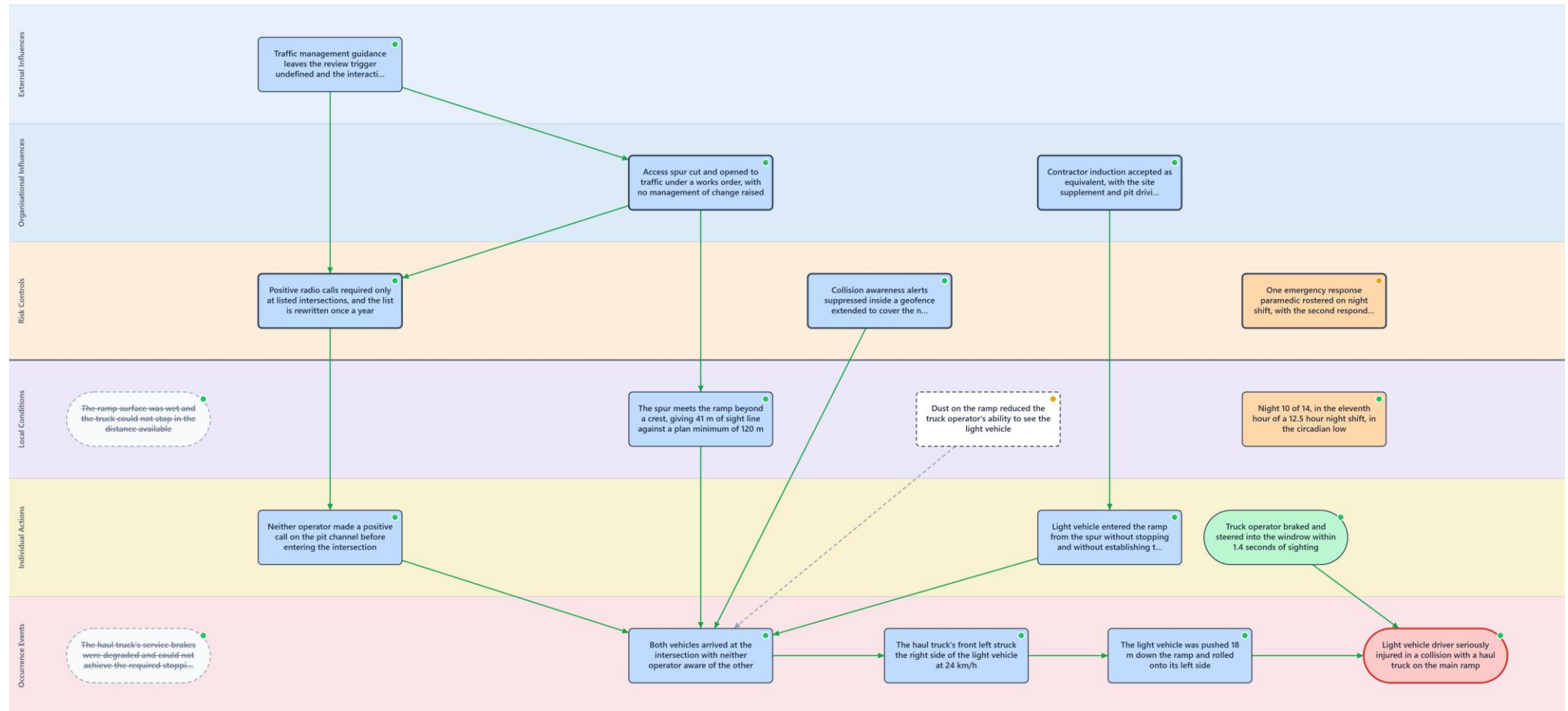
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AcciMap

Haul truck and light vehicle collision, Bindarra Gold Mine

Rendered 04/10/2026, 12:25:16 pm



Existence: did it exist or occur? Influence: did it influence this occurrence? Importance: is it worth addressing regardless?

External Influences

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	MSG-14 clause 3.2 requires the traffic management plan to be reviewed within twelve months and after any significant change to the road network, and defines neither the term nor an example of one. Clause 5.1 requires the operator to nominate the intersections and interaction zones at which positive communication is required, and sets no minimum and no basis for the nomination. Clause 5.4 requires a light vehicle to be detectable by a heavy vehicle operator and offers five alternative means, any one of which satisfies it. Clause 3.5 provides that the plan is a site document, not approved by and not submitted to the Inspectorate except on request after an occurrence. Bindarra's plan met the guideline on its face on the day of the occurrence: it had been reviewed inside twelve months, it nominated nine interaction zones, and it provided for detectability by line of sight and by elevated marking. The mine was not in breach.
Influence	Pass - likely (70%)	Three separate terms in the one guideline are left to the site to fill in, and at Bindarra all three were filled in by the same document, reviewed once a year, with no outside eye on any of it.
Importance	Not tested	(none)

Investigator notes: The mine is not in breach of anything, which is exactly why this belongs in External Influences rather than in a compliance finding. Be careful in the report not to let this read as blaming the regulator; the point is the shape of the discretion, not a failure by the Inspectorate.

Evidence attached

Issued by the State Mines Inspectorate, revision 4, 2021. A guideline made under regulation 6.22 of the Mining Operations Safety Regulations 2018. Compliance with the guideline is one way of discharging the regulation; the regulation itself requires only that "the operator shall manage the risks arising from the interaction of vehicles at the mine". > ****3.2**** The operator shall prepare and maintain a traffic management plan > for the mine. The plan shall be reviewed at intervals not exceeding twelve > months, and following any significant change to the mine's road network, > vehicle fleet or operating pattern. > > ****3.5**** The plan is a site document. It is not approved by, and need not > be submitted to, the Inspectorate, except on request following an > occurrence or an inspection. > > ****5.1**** The operator shall nominate the intersections, crossings and other > interaction zones at the mine at which positive communication between > vehicles is required before entry, and shall record them in the plan. > > ****5.4**** Where a light vehicle and a heavy vehicle may interact, the > operator shall provide for the light vehicle to be detectable by the heavy > vehicle operator. Detectability may be provided by line of sight, > elevated marking, radio communication, a proximity detection system, or a > combination of these. The guideline does not define "significant change" at 3.2, and contains no example of one. It does not say who nominates the interaction zones at 5.1 or on what basis, and sets no minimum. It does not say which of the five means at 5.4 must be provided, or how many.

Source: Mines Safety Guideline MSG-14 revision 4, made under Mining Operations Safety Regulations 2018 r 6.22

Owner: Mining Manager, Corella Resources. Issued 12/11/2025. Next review due 12/11/2026. > ****4.1**** Light vehicles shall give way to heavy vehicles at all times and > in all circumstances. > > ****4.3.1**** A positive call shall be made on the relevant pit channel before > entering any intersection listed at Appendix B, and entry shall not be > made until the call is acknowledged or

the channel is confirmed clear. > > ****4.3.2**** At intersections not listed at Appendix B, vehicles shall give > way in accordance with clause 4.1. No call is required. > > ****6.2**** At every intersection, the available sight distance shall be not > less than the stopping distance of the largest vehicle using the road at > the posted speed, and in no case less than 120 m. Where this cannot be > achieved, the intersection shall be controlled by a stop control, > signalised control or traffic controller. > > ****6.4**** Every intersection shown on the road layout drawing at Schedule 1 > shall be provided with an intersection warning sign 50 m in advance on > each approach, a give-way control on the minor road, and delineation at > the mouth. > > ****9.1**** This plan shall be reviewed annually, and following any > significant change to the mine's road network. > ****Appendix B - Intersections requiring positive communication**** > > Nine intersections are listed: the ROM pad turn-off, the RL 360 crossover, > the Stage 2 ramp junction, the Stage 2 dump access, the Stage 3 ramp > junction, the workshop apron entry, the explosives magazine access, the > raw water pump station access and the light vehicle parking bay at RL 400. > > ****Note to Appendix B.**** This appendix is reviewed and reissued at the > annual review of this plan (clause 9.1). An intersection is added to this > appendix where it appears on the road layout drawing at Schedule 1 and the > review assesses it as requiring positive communication. The road layout drawing at Schedule 1 is dated 12/11/2025 and is reissued at the annual review. The RL 312 dewatering spur does not appear on it. No provision anywhere in BIN-TMP-004 allows an intersection to be added to Appendix B, or to Schedule 1, between annual reviews, and no provision requires a new road to be referred to the plan's owner when it is cut. Appendix B is reproduced on a laminated board in each crib room. The board in the Stage 4 crib room is the 12/11/2025 issue.

Source: BIN-TMP-004 revision 7, issued 12/11/2025, next review due 12/11/2026

Document register, BIN-TMP-004. Annual reviews completed 14/11/2023, 10/11/2024 and 12/11/2025, each reissuing Schedule 1 and Appendix B. No out-of-cycle review has ever been carried out under clause 9.1, in the plan's eleven year history. The survey department's road register records 14 new roads cut in the Stage 4 pit since the 12/11/2025 review, of which 11 were cut under a works order and 3 under a management of change record. The three with a management of change record were referred to the plan's owner. None of the 11 was. ****Hazard report system extract.**** Corella Resources reporting system, search of all reports raised at Bindarra in the 24 months to 11/08/2026 containing the terms sight distance, blind, intersection, give way or light vehicle. Four reports returned, including HAZ-2026-0917. The other three, from 03/2025, 09/2025 and 02/2026, each describe a light vehicle and a haul truck meeting unexpectedly at an intersection not listed at Appendix B, at three different locations. All four were closed by the department that received them, within three days, with an entry recording that the crew had been reminded to take care and that heavy vehicles have right of way. None was referred to the traffic management plan owner. None resulted in a change to Appendix B, to Schedule 1, or to any physical control.

Source: Document register BIN-TMP-004; Corella reporting system search, 24 months to 11/08/2026

Organisational Influences

Access spur cut and opened to traffic under a works order, with no management of change raised

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	CRL-PRO-022 clause 5.2 requires a management of change record before any change to the mine's road network. Clause 5.3 excludes routine earthworks carried out under a works order within an approved pit design where no new plant is introduced. WO-26-4471 cut 240 m of access to bore DW-14 at RL 312, classified the work as routine earthworks within the approved Stage 4 design, recorded management of change as not required under clause 5.3, and left its handover field reading N/A. The three referrals that a management of change record would have triggered at clause 5.4 are the traffic management plan owner, the survey department for the fleet management system road network, and the contractor's technical services group for the collision awareness system geofences. None of the three was made. The in-cab road network published on 24/07/2026 was still current on the night and shows the main ramp with no intersection at RL 312. Of the fourteen new roads cut in Stage 4 since the last plan review, the three raised under a management of change record were referred to the plan's owner and none of the eleven raised under a works order was.
Influence	Pass - very likely (90%)	A works order built an intersection into a haul road and the classification on its front page removed every consequence of having done so. Eleven roads have been built the same way since the last review.
Importance	Not tested	(none)

Investigator notes: The clearest single organisational finding in the investigation. Everything that should have followed a new road into an operating pit lives inside the management of change record, and the exclusion at clause 5.3 removed the record.

Evidence attached

****Haul truck HT-407, descending the main ramp, empty.**** | Time (UTC) | Speed | Position | Note |
 | --- | --- | --- | --- | | 18:45:58 | 41 km/h | RL 368, main ramp | Second cycle after the crib break;
 tray down, empty | | 18:46:31 | 39 km/h | RL 348 | Retarder on the second position, steady descent
 | | 18:47:06 | 37 km/h | RL 314, 41 m above the spur intersection | Last data point before the brake
 application | | 18:47:07.4 | 37 km/h | RL 313 | Service brake pressure rises from 0 to 620 kPa in 0.4
 s; retarder to maximum; steering input 14 degrees right | | 18:47:09.2 | 32 km/h | RL 313 | CAS
 proximity event logged (see below); no alert raised in cab | | 18:47:12.0 | 24 km/h | RL 312, spur
 intersection | Deceleration spike; collision | | 18:47:14.3 | 0 km/h | 19 m below the intersection |
 Vehicle stationary, park brake applied | Braking analysis: 1.4 seconds elapsed between the last
 steady-state data point and the first brake pressure rise, which is the operator's response time on
 this download. Mean deceleration from the brake application to the collision was 1.85 m/s²,
 against 1.9 m/s² recorded for this unit in the monthly brake performance test on 03/08/2026. The
 retarder and the service brakes performed as specified throughout. Had no braking been applied
 the collision speed would have been 37 km/h rather than 24 km/h. Road network: the FMS in-cab
 map for the Stage 4 pit was last published on 24/07/2026. It does not contain the RL 312
 dewatering spur, which was cut on 31/07/2026. The in-cab display therefore showed the main
 ramp with no intersection at RL 312 on the night of the occurrence. Publishing a revised road
 network is a survey task raised by a management of change record; no such record exists for the
 spur. ****Light vehicle LV-118.**** LV-118 is not dispatched and carries no FMS. Its CAS transponder

is a passive tag. Its movements are known only from the CAS proximity log, the driver's account and the scene evidence.

Source: Warrigal Mining Services technical services report WMS-TS-2026-114, issued 18/08/2026

> ****5.2**** A management of change record shall be raised before any of the > following: a change to the mine's road network; the introduction of a new > vehicle type; a change to a pit design; a change to a documented control. > > ****5.3**** Clause 5.2 does not apply to routine earthworks carried out under > a works order within an approved pit design, where no new plant or vehicle > type is introduced. Such work is managed under the works order. > > ****5.4**** Where a management of change record is raised, the change owner > shall complete the impact assessment at Schedule 2, which includes > referral to the traffic management plan owner, the survey department for > the fleet management system road network, and the contractor's technical > services group for the collision awareness system configuration. The three referrals at 5.4 exist only inside the management of change record. The works order form at CRL-PRO-022 Schedule 4 has no equivalent fields, and no referral of any kind.

Source: CRL-PRO-022 revision 3, clauses 5.2 to 5.4 and Schedule 4

> ****Scope:**** Cut and form 240 m of access road to dewatering bore DW-14 at > RL 312, Stage 4 pit, to a formed width of 8 m with windrows to both sides. > > ****Justification:**** Access required for the commissioning and servicing of > DW-14 from 02/08/2026. > > ****Change classification:**** Routine earthworks within the approved Stage 4 > design. No new plant or vehicle type. Management of change not required, > CRL-PRO-022 clause 5.3. > > ****Raised by:**** Mining Engineer, 28/07/2026. ****Approved by:**** Mining > Superintendent, 29/07/2026. > > ****Works completed:**** 31/07/2026. > > ****Handover:**** N/A. The works order carries no scope for signage, no survey pickup of the new intersection, no referral to the traffic management plan, no request to publish a revised fleet management system road network, and no review of the collision awareness system geofences. The handover field is the only place any of them could have been recorded and it reads "N/A".

Source: Works order WO-26-4471, raised 28/07/2026, completed 31/07/2026

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

Test	Result	Supporting text
Existence	Pass - extremely likely (95%)	CRL-STD-009 clause 7.3 allows a contractor's own general induction to be accepted as equivalent where the contractor holds a current recognition agreement, and Merrivale Survey & Geotechnical has held one since 2023. Clause 7.4 states that equivalence does not extend to the Bindarra site-specific supplement or to the Bindarra light vehicle pit driving assessment, which shall be completed at site by every person who will drive in an operating pit and shall not be substituted. Clause 7.5 makes the engaging department responsible for verifying 7.4 before a card is issued. The gatehouse issues cards against the recognition agreement alone, holds no record of either item, and is not asked for one; nothing in the access card system distinguishes a person who may drive in an operating pit from a person who may not. The driver's training file records the Merrivale online module of 24/07/2026 and a 20 minute gatehouse briefing, and no supplement and no pit driving assessment. He had never seen BIN-TMP-004 and did not know it existed.
Influence	Pass - likely (80%)	A person was driving a light vehicle on an active haul road on his third night at the mine, without the assessment the site's own standard says may not be substituted, and no part of the system that let him through was looking for it.
Importance	Not tested	(none)

Investigator notes: Watch the wording of this in the report. The standard is right and says the right thing; what failed is that the only place the verification could have happened has no field for it. Do not write it as a failure by the driver.

Evidence attached

> ****7.3**** A contractor's own general induction may be accepted as equivalent > to the Bindarra general induction where the contractor holds a current > recognition agreement with Corella Resources. > > ****7.4**** Equivalence under 7.3 does not extend to the Bindarra > site-specific supplement, or to the Bindarra light vehicle pit driving > assessment, which shall be completed at site by every person who will > drive a vehicle in an operating pit and shall not be substituted. > > ****7.5**** The engaging department is responsible for verifying that 7.4 has > been completed before a site access card is issued. Merrivale Survey & Geotechnical has held a recognition agreement since 2023. The gatehouse issues site access cards against the recognition agreement alone; it holds no record of the supplement or the pit driving assessment and is not asked for one. Nothing in the access card system distinguishes a person who may drive in an operating pit from a person who may not.

Source: CRL-STD-009 revision 5, clauses 7.3 to 7.5

Fourteen months at Bindarra, seven years on haul trucks across three sites. Verification of competency on the 220 t class unit issued 06/2025 and current. Annual refresher completed 02/2026. Site general induction 06/2025. The record shows no traffic management refresher of any kind since the general induction, and the general induction module on traffic management is a 23 slide package that names the nine Appendix B intersections and does not mention what to do at an intersection that is not listed. There is no training record anywhere on site, for any person, covering the collision awareness system's suppression geofences. ****Light vehicle driver, induction and training record.**** Merrivale Survey & Geotechnical training file and the Bindarra gatehouse access card register. Merrivale general mining induction, a two hour online module, completed 24/07/2026. Bindarra site access card issued 09/08/2026 against Merrivale's

recognition agreement, following a 20 minute gatehouse briefing covering muster points, speed limits and personal protective equipment. There is no record of the Bindarra site-specific supplement being issued, completed or signed for. There is no record of the Bindarra light vehicle pit driving assessment. Neither record is required by the gatehouse before a card is issued, and neither was asked for. The driver had worked three night shifts at Bindarra: 09/08, 10/08 and 11/08.

Source: Warrigal training file, Merrivale training file and the Bindarra gatehouse access card register

Risk Controls

Positive radio calls required only at listed intersections, and the list is rewritten once a year

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	BIN-TMP-004 clause 4.3.1 requires a positive call on the pit channel before entering any intersection listed at Appendix B, and 4.3.2 requires no call anywhere else. Appendix B lists nine intersections. Its own note provides that the appendix is reviewed and reissued at the annual review, and that an intersection is added where it appears on the road layout drawing at Schedule 1 and the review assesses it as requiring positive communication. Schedule 1 is dated 12/11/2025 and is reissued at the annual review. No provision anywhere in the plan allows an intersection to be added between reviews, and no provision requires a new road to be referred to the plan's owner when it is cut. The RL 312 spur is on neither document. Eleven of the fourteen new Stage 4 roads since the last review were never referred, and no out-of-cycle review has been held in the plan's eleven year history. The truck operator, the light vehicle driver and the night shift supervisor each described the same practice independently: calls are made at the nine intersections on the crib room board and nowhere else.
Influence	Pass - very likely (90%)	This is the mine's own defence against exactly this interaction, and at this intersection it was unarmed by its own terms. The next date on which the spur could have been listed is 12/11/2026, three months after the collision.
Importance	Not tested	(none)

Investigator notes: The safety issue the map exists to find. The control is sound and its trigger is circular: an intersection reaches the list through a drawing that is only reissued at the review, and nothing requires a new road to be referred at all. Recommend this one in its own right whatever else the report says.

Evidence attached

18:22:41 Dispatch All Stage 4 units, crib's done, back to work please. 18:22:49 HT-411 411 rolling. 18:22:53 HT-407 407 rolling, first load. 18:31:18 HT-407 407 loaded, heading up. 18:33:02 Excavator 2 Dig's getting tight on the north side, I'll need the dozer before the next pass. 18:33:14 Dispatch Copy, I'll task 62 when he's free. 18:38:55 HT-407 407 dumped, coming back down. 18:41:30 HT-411 411 loaded. 18:46:12 Excavator 2 Who've I got next? 18:46:18 Dispatch 407's about four minutes off you. 18:47:41 HT-407 [raised voice] Dispatch, 407. I've hit a light vehicle on the ramp at three-twelve. I've hit a light vehicle. He's on his side. 18:47:49 Dispatch 407, say again? 18:47:52 HT-407 I've collided with a light vehicle at the three-twelve spur. He's rolled. I need ERT and I need them now. 18:48:01 Dispatch All Stage 4 units stop. All units stop. Emergency on the pit channel. 407, are you out of your truck? 18:48:09 HT-407 I'm getting out now. 18:48:14 Dispatch Negative 407, stay in the cab until I've stopped the traffic. Give me thirty seconds. 18:48:44 Dispatch 407, ramp's clear above and below you, you're safe to get out. ERT are rolling from the surface. 18:49:02 Supervisor Supervisor's en route from the ROM pad, five minutes. 407, is he conscious? 18:49:30 HT-407 He's conscious. He's talking to me. He can't get out, the door's against the ground. 18:49:38 Supervisor Don't move him and don't try to get him out. Wait for ERT. 18:51:20 ERT 1 ERT 1 entering the pit, RL three-eighty.

Source: RAD-2026-0332, transcribed 17/08/2026, verified by a second investigator

Document register, BIN-TMP-004. Annual reviews completed 14/11/2023, 10/11/2024 and 12/11/2025, each reissuing Schedule 1 and Appendix B. No out-of-cycle review has ever been carried out under clause 9.1, in the plan's eleven year history. The survey department's road register records 14 new roads cut in the Stage 4 pit since the 12/11/2025 review, of which 11 were cut under a works order and 3 under a management of change record. The three with a management of change record were referred to the plan's owner. None of the 11 was. ****Hazard report system extract.**** Corella Resources reporting system, search of all reports raised at Bindarra in the 24 months to 11/08/2026 containing the terms sight distance, blind, intersection, give way or light vehicle. Four reports returned, including HAZ-2026-0917. The other three, from 03/2025, 09/2025 and 02/2026, each describe a light vehicle and a haul truck meeting unexpectedly at an intersection not listed at Appendix B, at three different locations. All four were closed by the department that received them, within three days, with an entry recording that the crew had been reminded to take care and that heavy vehicles have right of way. None was referred to the traffic management plan owner. None resulted in a change to Appendix B, to Schedule 1, or to any physical control.

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Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The collision awareness system on the Stage 4 haul fleet operates at Level 7, operator advisory; Level 9 intervention is available on the fleet and is not enabled at Bindarra. Advisory alerts are raised on a tag entering the 60 m ring above a 10 km/h closing rate, except inside the S4-LOAD suppression geofence, which was drawn around the loading face on 14/03/2025 to stop nuisance alerts and extended northward on 09/06/2026 when the face advanced. The extended boundary lies 180 m up-ramp of the RL 312 intersection, placing the intersection inside the suppressed area. The extension was made as a system configuration task by the contractor's technical services group, is not recorded in the traffic management plan, and was not reviewed against it. The log records a proximity event at 02:47:09.2 (18:47:09.2 UTC) at 21.4 m and 13 km/h closing, alert suppressed. In the twelve months before the occurrence HT-407 raised 412 advisory alerts, every one outside the geofence and none inside it, while 68 proximity events inside it were suppressed, nine of them a light vehicle tag on the main ramp below the boundary. No training record anywhere on site, for any person, covers the geofences.
Influence	Pass - likely (75%)	Nine earlier events of exactly this kind were recorded by the system, visible to anyone who queried it, and reported to nobody, because the standing report covers alert counts and system health and does not report suppressions at all.
Importance	Not tested	(none)

Investigator notes: Both the operator and the supervisor raised this unprompted and the statistics bear them out. The finding is not that the geofence exists; nuisance alerts around an excavator will destroy a system. The finding is that its boundary moved 180 m in June and nobody outside the group that moved it was told.

Evidence attached

****Configuration as found on HT-407.**** System operating at Level 7 (operator advisory). Level 9 (automated intervention) is available on this fleet and is not enabled at Bindarra. Advisory alerts are raised on a vehicle tag entering the 60 m detection ring at a closing rate above 10 km/h, except inside a suppression geofence. ****Suppression geofence "S4-LOAD".**** Drawn 14/03/2025 around the Stage 4 loading face to stop nuisance alerts between the excavator, the trucks under it and the service vehicles working in the same area, where the alert rate made the system unusable. Extended northward on 09/06/2026 when the loading face advanced. The extended geofence boundary lies 180 m up-ramp of the RL 312 spur intersection, so the intersection sits inside the suppressed area. The extension was made by the contractor's technical services group as a system configuration task. It is not recorded in the traffic management plan and was not reviewed against it. ****Event log, 11/08/2026.**** `` 18:47:09.200 PROXIMITY HT-407 tag LV-118 range 21.4 m closing 13 km/h zone S4-LOAD alert SUPPRESSED (geofence) 18:47:12.000 IMPACT HT-407 longitudinal 1.9 g logged by accelerometer 18:47:12.100 PROXIMITY HT-407 tag LV-118 range 0.0 m zone S4-LOAD alert SUPPRESSED (geofence) `` The proximity event at 18:47:09.2 was logged 2.8 seconds before the collision, 1.8 seconds after the operator had already begun braking. Outside the geofence the same event would have raised an advisory tone and a screen indication in the cab.

Source: CAS configuration and event log, HT-407, extracted 12/08/2026, WMS-TS-2026-114

Fourteen months at Bindarra, seven years on haul trucks across three sites. Verification of competency on the 220 t class unit issued 06/2025 and current. Annual refresher completed

02/2026. Site general induction 06/2025. The record shows no traffic management refresher of any kind since the general induction, and the general induction module on traffic management is a 23 slide package that names the nine Appendix B intersections and does not mention what to do at an intersection that is not listed. There is no training record anywhere on site, for any person, covering the collision awareness system's suppression geofences. **Light vehicle driver, induction and training record.** Merrivale Survey & Geotechnical training file and the Bindarra gatehouse access card register. Merrivale general mining induction, a two hour online module, completed 24/07/2026. Bindarra site access card issued 09/08/2026 against Merrivale's recognition agreement, following a 20 minute gatehouse briefing covering muster points, speed limits and personal protective equipment. There is no record of the Bindarra site-specific supplement being issued, completed or signed for. There is no record of the Bindarra light vehicle pit driving assessment. Neither record is required by the gatehouse before a card is issued, and neither was asked for. The driver had worked three night shifts at Bindarra: 09/08, 10/08 and 11/08.

Source: Warrigal training file, Merrivale training file and the Bindarra gatehouse access card register

Warrigal Mining Services technical services, extract of the fleet CAS event log for the 12 months to 11/08/2026. HT-407 raised 412 operator advisory alerts in that period, every one of them outside the S4-LOAD geofence, and none inside it. Across the Stage 4 haul fleet, 68 proximity events were logged inside the S4-LOAD geofence and suppressed. Nine of those 68 involved a light vehicle tag on the main ramp below the geofence boundary, the same configuration of vehicles and location as the occurrence. The suppressed events are recorded in the system and are visible to anyone who queries it. Nobody had queried it: the technical services group's standing report covers alert counts and system health, and does not report suppressed events at all.

Source: Warrigal technical services fleet CAS event log extract, 12 months to 11/08/2026

One emergency response paramedic rostered on night shift, with the second responder called from the village

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The Bindarra night shift is rostered with one qualified mines rescue paramedic, based at the surface medical facility. The second responder, who must be present before any extrication is begun, is called in from the accommodation village 11 km from the gatehouse. On the night, dispatch paged the team at 02:48, the first response vehicle entered the pit at 02:51 and reached the scene at 02:54, the second responder arrived with the hydraulic rescue tool at 03:07, and the driver was extricated at 03:14, 27 minutes after the collision. The post-incident review found the extrication delayed by approximately 9 minutes waiting for the second responder and the tool, and recorded that the delay did not affect the clinical outcome because the injuries were stable and the patient was conscious throughout.
Influence	Fail - unlikely (10%)	A nine minute wait for the second half of an extrication capability is a defence that is not there when it is needed, and nothing about the arrangement changes with the severity of the casualty it is called to.
Importance	Pass - very likely (90%)	It did not influence this occurrence and it is still a finding. The driver was conscious, stable, restrained and protected by an intact roll-over structure, and the delay cost him nothing. Change one thing, a fuel fire, an unconscious casualty, an airway that needed the vehicle opened in the first ten minutes, and the same nine minutes is a different outcome. The gap that matters is structural rather than clinical: the site's extrication capability on night shift is a single person plus a drive from the village, and no part of the arrangement scales with what has happened.

Investigator notes: Left in draft deliberately. The question of what the roster looks like across the other three night shifts on site is out with the emergency response coordinator and this object should not be finalised until that answer is in, even though its own tests are settled.

Evidence attached

Bindarra emergency response log and the post-incident review, obtained 26/08/2026. The night shift is rostered with one qualified mines rescue paramedic, who was at the surface medical facility. The second responder, required before any extrication, is called in from the accommodation village 11 km from the gatehouse. Dispatch paged the emergency response team at 02:48 local. The first response vehicle entered the pit at 02:51 and reached the scene at 02:54. The second responder arrived on scene at 03:07 with the hydraulic rescue tool. The driver was extricated through the rear of the vehicle at 03:14, 27 minutes after the collision, and reached the surface medical facility at 03:41. The post-incident review found the extrication was delayed by approximately 9 minutes waiting for the second responder and the tool, and recorded that the delay did not affect the clinical outcome on this occasion because the injuries were stable and the patient was conscious throughout.

Source: Bindarra emergency response log and post-incident review, obtained 26/08/2026

Local Conditions

The ramp surface was wet and the truck could not stop in the distance available

Test	Result	Supporting text
Existence	Fail - very unlikely (5%)	Hypothesis tested because an 18 m push down a 9 per cent grade invites a question about the surface. It fails. The met station records no precipitation on the night and the last rainfall at the site was 26 days before the occurrence. The water cart log puts the last pass on that section at 01:10 (17:10 UTC), 1 hour 37 minutes before, and the standing night shift instruction is a pass every 90 minutes, so the surface had been drying for longer than the interval rather than freshly watered. The investigator examined the ramp at 06:20 local and found it dry throughout, with no damp patches, ponding or greasy film at or above the intersection. The braked tyre scuff runs continuously for 62 m with no sign of a locked or sliding wheel, and the download gives a mean deceleration of 1.85 m/s ² against the unit's own tested 1.9 m/s ² on a dry 9 per cent grade. There is no evidence of a wet or slippery surface from any source.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: Raised early because a 220 tonne truck sliding down a wet ramp is the first thing anybody pictures. Disposed of in a morning and kept on the map so the report can say it was tested rather than assumed away.

Evidence attached

The main ramp at RL 312 runs approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows, surface dry and well maintained, no corrugation or loose material beyond the normal running surface. The RL 312 dewatering spur joins it from the east on the inside of a left-hand bend, at an included angle of 71 degrees. Marks recovered, measured from the eastern edge of the spur mouth: | Mark | Detail | | --- | --- | | HT-407 left front tyre, braked | Continuous scuff commencing 38.5 m up-ramp of the spur mouth and running 62 m to the truck's resting position | | HT-407 steering | The scuff deviates 1.4 m to the right over its first 20 m, consistent with the 14 degree steering input on the download | | Impact position | 6.2 m into the ramp from the eastern edge of the spur mouth, 3.1 m out from the inner windrow toe | | LV-118 gouge | Continuous gouge from the impact position running 18.4 m down-ramp, changing at 11.2 m from tyre scuff to body panel gouge, which is where the vehicle went over | | LV-118 rest position | On its left side, against the inner windrow, 18.4 m down-ramp of the impact position, facing down-ramp | | Debris | Right-side glass and mirror fragments from 4.8 m to 14 m down-ramp of the impact position | LV-118 damage is concentrated on the right side from the B pillar forward. The roll-over protection structure is deformed inward at the right rear mounting but was not breached. The driver's seatbelt was found fastened and the webbing showed loading marks. The 4.5 m amber beacon whip was sheared at its base and recovered 9 m from the vehicle; its lamp was tested after recovery and worked. HT-407 damage is confined to the front left bumper, the left mudguard and the lower two steps of the access ladder. There is no damage above the bumper line.

Source: INV-2026-0332-SCENE, scene examined 12/08/2026 before recovery

****Light.**** The occurrence was at 02:47 local, in darkness. There is no fixed lighting anywhere in the Stage 4 pit; the only lighting is vehicle lighting and the light plants at the loading face, 1.1 km away and out of sight behind the crest. Moonset was at 23:18 local the previous evening.

****Surface and weather.**** Bindarra met station, ROM pad: wind south-east 8 km/h, temperature 9 degrees, no precipitation. The last recorded rainfall at the site was 26 days before the occurrence. The investigator examined the ramp surface at 06:20 local and found it dry throughout, with no damp patches, ponding or greasy film at or above the intersection. ****Water cart and dust.**** The water cart log records the main ramp between RL 280 and RL 400 last watered at 01:10 local (17:10 UTC), 1 hour 37 minutes before the occurrence. The standing instruction on night shift is a ramp pass every 90 minutes, so a pass was 7 minutes overdue. The fixed dust monitor at the ramp crest, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre in the 02:40 to 02:50 local interval (18:40 to 18:50 UTC), against a shift mean of 88 and a site trigger of 250. That is a measure of airborne particulate at the monitor head. Nothing at Bindarra measures visibility, and no measurement exists of what could be seen from either cab on the night.

Source: Met station, water cart log and fixed dust monitor records, night of 11 to 12/08/2026

The spur meets the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The survey pickup of 13/08/2026 places the spur mouth 34 m beyond the high point of a crest in the ramp. An operator descending at an eye height of 5.80 m first has line of sight to the spur mouth at 41 m; a driver at the spur mouth at an eye height of 1.45 m first has line of sight to a descending truck at 38 m, with the 1.6 m inner windrow taking the truck's lower body out of view for the last 25 m of the spur's approach. BIN-TMP-004 clause 6.2 requires available sight distance at every intersection to be not less than the stopping distance of the largest vehicle at the posted speed and in no case less than 120 m, and requires a stop, signalised or controlled intersection where that cannot be achieved. The stopping distance of a 220 t class truck empty at the posted 40 km/h on a 9 per cent descending grade, on the manufacturer's data with a 1.5 second response, is 96 m. The intersection meets neither limb of clause 6.2. It carried no warning sign, no give-way control, no delineation and no posted speed, because clause 6.4 applies only to intersections shown on the road layout drawing and this one is not on it. Seven days earlier a dewatering technician reported the same sight line from the same position in his own words.
Influence	Pass - very likely (90%)	41 m against a 96 m stopping distance means that a truck at the posted speed cannot stop for anything at this intersection, whatever the operator does and however alert he is.
Importance	Not tested	(none)

Investigator notes: The measurement the organisational half of the map rests on. Make sure the report draws the line between this number and the stopping distance: at 41 m, attention, training and a radio call are all downstream of a geometry that cannot be driven safely.

Evidence attached

The survey department picked up the intersection, the crest and the eye positions of both vehicles. The spur meets the ramp 34 m beyond the high point of a crest in the ramp. An operator descending the ramp, seated at an eye height of 5.80 m above the running surface, first has line of sight to the spur mouth at 41 m. A driver at the spur mouth, at an eye height of 1.45 m, first has line of sight to a descending truck at 38 m; the inner windrow, 1.6 m high, stands between the spur and the ramp for the last 25 m of the spur's approach and takes the truck's lower body out of view before that. The Bindarra traffic management plan, clause 6.2, requires that at every intersection the available sight distance shall be not less than the stopping distance of the largest vehicle using the road at the posted speed, and in no case less than 120 m. The stopping distance of a 220 t class truck travelling empty at the posted 40 km/h on a 9 per cent descending grade, calculated with the manufacturer's data and a 1.5 second response, is 96 m. The available sight distance at this intersection is 41 m. The intersection does not meet clause 6.2 on either limb.

Source: Mine survey pickup SUR-26-1188, 13/08/2026, at the investigation's request

At the time of the occurrence the RL 312 spur intersection carried: - no intersection warning sign on either ramp approach; - no give-way or stop control at the spur mouth; - no delineation, reflective marker posts or windrow gap markers; - no posted speed for the spur itself. A single "ROAD CLOSED - CONSTRUCTION" sign had stood at the spur mouth while the road was being cut. The works log records it removed on 02/08/2026, the day the bore was energised and the spur was first used by light vehicles. Nothing was installed in its place. The traffic management plan, clause 6.4, requires an intersection warning sign 50 m in advance on each approach, a give-

way control on the minor road, and delineation at the mouth. Clause 6.4 applies to intersections shown on the plan's own road layout drawing. The RL 312 spur is not on that drawing.

Source: INV-2026-0332-SCENE, signage and delineation, 12/08/2026

****Submitted by:**** Dewatering technician, Corella Resources ****Submitted:**** 04/08/2026 23:48 local, the same night as the near miss it describes and seven days before the occurrence
****Category:**** Hazard - traffic and mobile equipment ****Risk rating applied by the submitter:**** Medium ****Risk rating on closure:**** Low ****Status:**** Closed 06/08/2026 ****Text as submitted:**** ---
Coming out of the new dewatering road onto the main ramp tonight about ten past ten I have come out and there was a truck coming down the ramp maybe sixty metres up. He has not seen me and I have not seen him until I was already out on the road. I have reversed back into the road and let him go past. Nothing happened but it was close enough that I am writing it up. The problem is you cannot see up the ramp from that road. There is a rise in the ramp and a windrow along the side and you are out on the road before you can see anything. There is no sign on the ramp telling the trucks the road is there and there is no give way on our side. The closed sign that was there while they were building it has been taken away. The trucks are doing forty coming down empty. I do not think they know the road is there. --- ****Closure entry, made 06/08/2026 by the Dewatering Superintendent:**** "Spoke with the dewatering crew at pre-start 05/08 and reminded them to take care entering the main ramp and to give way to haul trucks at all times. Crew acknowledged. Trucks have right of way on the ramp per the traffic management plan. No further action required. Closed." The report was not referred to the mining department, the contractor, the survey department or the traffic management plan owner. The closure entry is the only action recorded against it.

Source: Hazard report HAZ-2026-0917, submitted 04/08/2026, closed 06/08/2026

Dust on the ramp reduced the truck operator's ability to see the light vehicle

Test	Result	Supporting text
Existence	Inconclusive - about as likely as not (45%)	For: the water cart's last pass on that section was 1 hour 37 minutes before the occurrence against a standing 90 minute interval, so a pass was 7 minutes overdue. The fixed dust monitor at the ramp crest, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre in the ten minutes covering the occurrence against a shift mean of 88. The truck operator described dust hanging on the ramp, said the cart had been through earlier and it dries out, and the descent was behind his own lights into an unlit pit, which is the condition in which airborne dust is worst for a driver.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: Left in draft on purpose. Two honest accounts that do not agree, one measurement that shows a deviation and not an effect, and no way to close it with what exists. Say so in the report rather than picking a side.

Evidence attached

The main ramp at RL 312 runs approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows, surface dry and well maintained, no corrugation or loose material beyond the normal running surface. The RL 312 dewatering spur joins it from the east on the inside of a left-hand bend, at an included angle of 71 degrees. Marks recovered, measured from the eastern edge of the spur mouth: | Mark | Detail | | --- | --- | | HT-407 left front tyre, braked | Continuous scuff commencing 38.5 m up-ramp of the spur mouth and running 62 m to the truck's resting position | | HT-407 steering | The scuff deviates 1.4 m to the right over its first 20 m, consistent with the 14 degree steering input on the download | | Impact position | 6.2 m into the ramp from the eastern edge of the spur mouth, 3.1 m out from the inner windrow toe | | LV-118 gouge | Continuous gouge from the impact position running 18.4 m down-ramp, changing at 11.2 m from tyre scuff to body panel gouge, which is where the vehicle went over | | LV-118 rest position | On its left side, against the inner windrow, 18.4 m down-ramp of the impact position, facing down-ramp | | Debris | Right-side glass and mirror fragments from 4.8 m to 14 m down-ramp of the impact position | LV-118 damage is concentrated on the right side from the B pillar forward. The roll-over protection structure is deformed inward at the right rear mounting but was not breached. The driver's seatbelt was found fastened and the webbing showed loading marks. The 4.5 m amber beacon whip was sheared at its base and recovered 9 m from the vehicle; its lamp was tested after recovery and worked. HT-407 damage is confined to the front left bumper, the left mudguard and the lower two steps of the access ladder. There is no damage above the bumper line.

Source: INV-2026-0332-SCENE, scene examined 12/08/2026 before recovery

****Light.**** The occurrence was at 02:47 local, in darkness. There is no fixed lighting anywhere in the Stage 4 pit; the only lighting is vehicle lighting and the light plants at the loading face, 1.1 km away and out of sight behind the crest. Moonset was at 23:18 local the previous evening.

****Surface and weather.**** Bindarra met station, ROM pad: wind south-east 8 km/h, temperature 9 degrees, no precipitation. The last recorded rainfall at the site was 26 days before the occurrence. The investigator examined the ramp surface at 06:20 local and found it dry throughout, with no damp patches, ponding or greasy film at or above the intersection. ****Water cart and dust.**** The water cart log records the main ramp between RL 280 and RL 400 last watered at 01:10 local (17:10 UTC), 1 hour 37 minutes before the occurrence. The standing instruction on night shift is a ramp pass every 90 minutes, so a pass was 7 minutes overdue. The fixed dust monitor at the

ramp crest, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre in the 02:40 to 02:50 local interval (18:40 to 18:50 UTC), against a shift mean of 88 and a site trigger of 250. That is a measure of airborne particulate at the monitor head. Nothing at Bindarra measures visibility, and no measurement exists of what could be seen from either cab on the night.

Source: Met station, water cart log and fixed dust monitor records, night of 11 to 12/08/2026

Night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The operator was on night 10 of a 14 night swing, working 18:00 to 06:30 local, and the collision was at 02:47, 10 hours 47 minutes into the shift and inside the window of lowest circadian alertness. Every limit in the contractor's fatigue management standard was met: no more than 14 consecutive nights, a minimum 10 hour break, and no shift extended beyond 12 hours 40 minutes in the preceding nine. His own voluntary sleep record gives 6 hours 30 minutes in the day before and between 5 and 7 hours on each of the preceding nine days. He described himself in interview as tired in the way you are on night ten, not falling asleep, awake and driving properly.
Influence	Fail - unlikely (25%)	Ten consecutive nights at the bottom of the circadian cycle is a standing exposure built into the roster rather than a one-off, and it is worked by every operator on this pit on every swing.
Importance	Pass - likely (85%)	It failed Influence for this occurrence and remains a finding worth acting on. The Stage 4 night roster puts every haul truck operator at the wheel of a 220 tonne vehicle in the small hours of a tenth consecutive night, at a mine with no fixed lighting anywhere in the pit, where the sole defence against an unlit hazard at an unlisted intersection is the operator's own eyes. Nothing about that changes because it did not bite here, and the same roster is worked on three other pits on this site.

Investigator notes: The tempting object. It exists, it reads as a cause, and the evidence will not carry it for this occurrence. Recording it as an other risk factor is the honest answer and it is still going in the report.

Evidence attached

****Haul truck HT-407, descending the main ramp, empty.**** | Time (UTC) | Speed | Position | Note |
 | --- | --- | --- | --- | | 18:45:58 | 41 km/h | RL 368, main ramp | Second cycle after the crib break; tray down, empty | | 18:46:31 | 39 km/h | RL 348 | Retarder on the second position, steady descent | | 18:47:06 | 37 km/h | RL 314, 41 m above the spur intersection | Last data point before the brake application | | 18:47:07.4 | 37 km/h | RL 313 | Service brake pressure rises from 0 to 620 kPa in 0.4 s; retarder to maximum; steering input 14 degrees right | | 18:47:09.2 | 32 km/h | RL 313 | CAS proximity event logged (see below); no alert raised in cab | | 18:47:12.0 | 24 km/h | RL 312, spur intersection | Deceleration spike; collision | | 18:47:14.3 | 0 km/h | 19 m below the intersection | Vehicle stationary, park brake applied | Braking analysis: 1.4 seconds elapsed between the last steady-state data point and the first brake pressure rise, which is the operator's response time on this download. Mean deceleration from the brake application to the collision was 1.85 m/s², against 1.9 m/s² recorded for this unit in the monthly brake performance test on 03/08/2026. The retarder and the service brakes performed as specified throughout. Had no braking been applied the collision speed would have been 37 km/h rather than 24 km/h. Road network: the FMS in-cab map for the Stage 4 pit was last published on 24/07/2026. It does not contain the RL 312 dewatering spur, which was cut on 31/07/2026. The in-cab display therefore showed the main ramp with no intersection at RL 312 on the night of the occurrence. Publishing a revised road network is a survey task raised by a management of change record; no such record exists for the spur. ****Light vehicle LV-118.**** LV-118 is not dispatched and carries no FMS. Its CAS transponder is a passive tag. Its movements are known only from the CAS proximity log, the driver's account and the scene evidence.

Source: Warrigal Mining Services technical services report WMS-TS-2026-114, issued 18/08/2026

Warrigal Mining Services roster and site access records. A 14 night on, 7 off roster, working 18:00 to 06:30 local, a 12 hour 30 minute shift with a 30 minute crib. The occurrence was on night 10 of 14, 10 hours 47 minutes into the shift, at 02:47 local. Site access records confirm the previous nine shifts worked in sequence with no shift extended beyond 12 hours 40 minutes. The contractor's fatigue management standard sets a maximum of 14 consecutive night shifts and a minimum 10 hour break, both of which were met. The operator's own sleep record, kept voluntarily and provided by him, records 6 hours 30 minutes of sleep in the day before the occurrence and between 5 and 7 hours on each of the preceding nine days.

Source: Warrigal roster and site access records; the operator's own sleep record, provided voluntarily

Individual Actions

Neither operator made a positive call on the pit channel before entering the intersection

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The pit channel was logged continuously and the investigator listened to the whole period from the shift's return from crib to two minutes after the collision. There is no transmission by either vehicle about the RL 312 spur, the dewatering bore, or a light vehicle being anywhere in the Stage 4 pit. LV-118 made no transmission on the channel at all that shift beyond logging on at 18:04 (10:04 UTC), at the start of shift. No call was required: BIN-TMP-004 clause 4.3.2 requires none at an intersection not listed at Appendix B, and this one is not listed. The truck operator, the light vehicle driver and the night shift supervisor each described the same practice independently and without being led: calls are made at the nine intersections on the crib room board and at no others, and the supervisor said plainly that he has never corrected anybody for not calling elsewhere and that requiring calls everywhere would make the channel unusable.
Influence	Pass - likely (85%)	The one action that would have put each vehicle in the other's knowledge was available, cost nothing, and was on neither operator's mind, because no document put it there and the local practice matched the document exactly.
Importance	Not tested	(none)

Investigator notes: Careful with the wording of this one. It is an omission with no owner: the plan expressly requires no call there, every operator on the pit described the same practice, and the supervisor confirmed he has never pulled anybody up for it.

Evidence attached

18:22:41 Dispatch All Stage 4 units, crib's done, back to work please. 18:22:49 HT-411 411 rolling. 18:22:53 HT-407 407 rolling, first load. 18:31:18 HT-407 407 loaded, heading up. 18:33:02 Excavator 2 Dig's getting tight on the north side, I'll need the dozer before the next pass. 18:33:14 Dispatch Copy, I'll task 62 when he's free. 18:38:55 HT-407 407 dumped, coming back down. 18:41:30 HT-411 411 loaded. 18:46:12 Excavator 2 Who've I got next? 18:46:18 Dispatch 407's about four minutes off you. 18:47:41 HT-407 [raised voice] Dispatch, 407. I've hit a light vehicle on the ramp at three-twelve. I've hit a light vehicle. He's on his side. 18:47:49 Dispatch 407, say again? 18:47:52 HT-407 I've collided with a light vehicle at the three-twelve spur. He's rolled. I need ERT and I need them now. 18:48:01 Dispatch All Stage 4 units stop. All units stop. Emergency on the pit channel. 407, are you out of your truck? 18:48:09 HT-407 I'm getting out now. 18:48:14 Dispatch Negative 407, stay in the cab until I've stopped the traffic. Give me thirty seconds. 18:48:44 Dispatch 407, ramp's clear above and below you, you're safe to get out. ERT are rolling from the surface. 18:49:02 Supervisor Supervisor's en route from the ROM pad, five minutes. 407, is he conscious? 18:49:30 HT-407 He's conscious. He's talking to me. He can't get out, the door's against the ground. 18:49:38 Supervisor Don't move him and don't try to get him out. Wait for ERT. 18:51:20 ERT 1 ERT 1 entering the pit, RL three-eighty.

Source: RAD-2026-0332, transcribed 17/08/2026, verified by a second investigator

Owner: Mining Manager, Corella Resources. Issued 12/11/2025. Next review due 12/11/2026. >
 4.1 Light vehicles shall give way to heavy vehicles at all times and > in all circumstances. > >
 4.3.1 A positive call shall be made on the relevant pit channel before > entering any intersection listed at Appendix B, and entry shall not be > made until the call is acknowledged or

the channel is confirmed clear. > > ****4.3.2**** At intersections not listed at Appendix B, vehicles shall give > way in accordance with clause 4.1. No call is required. > > ****6.2**** At every intersection, the available sight distance shall be not > less than the stopping distance of the largest vehicle using the road at > the posted speed, and in no case less than 120 m. Where this cannot be > achieved, the intersection shall be controlled by a stop control, > signalised control or traffic controller. > > ****6.4**** Every intersection shown on the road layout drawing at Schedule 1 > shall be provided with an intersection warning sign 50 m in advance on > each approach, a give-way control on the minor road, and delineation at > the mouth. > > ****9.1**** This plan shall be reviewed annually, and following any > significant change to the mine's road network. > ****Appendix B - Intersections requiring positive communication**** > > Nine intersections are listed: the ROM pad turn-off, the RL 360 crossover, > the Stage 2 ramp junction, the Stage 2 dump access, the Stage 3 ramp > junction, the workshop apron entry, the explosives magazine access, the > raw water pump station access and the light vehicle parking bay at RL 400. > > ****Note to Appendix B.**** This appendix is reviewed and reissued at the > annual review of this plan (clause 9.1). An intersection is added to this > appendix where it appears on the road layout drawing at Schedule 1 and the > review assesses it as requiring positive communication. The road layout drawing at Schedule 1 is dated 12/11/2025 and is reissued at the annual review. The RL 312 dewatering spur does not appear on it. No provision anywhere in BIN-TMP-004 allows an intersection to be added to Appendix B, or to Schedule 1, between annual reviews, and no provision requires a new road to be referred to the plan's owner when it is cut. Appendix B is reproduced on a laminated board in each crib room. The board in the Stage 4 crib room is the 12/11/2025 issue.

Source: BIN-TMP-004 revision 7, issued 12/11/2025, next review due 12/11/2026

Light vehicle entered the ramp from the spur without stopping and without establishing the truck's position

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The driver's account, corrected by him on read-back from stopped to slowed, is that he came out of the spur at about 25 km/h, looked left up the ramp, saw nothing and drove out; he does not believe he came to a complete stop. The scene evidence is consistent: the impact is 6.2 m into the ramp from the eastern edge of the spur mouth, and the gouge runs from there. BIN-TMP-004 clause 4.1 requires light vehicles to give way to heavy vehicles at all times, which is the rule he described himself as applying. There was no give-way control, no stop line, no warning sign and no posted speed at the mouth, because clause 6.4 applies only to intersections on the road layout drawing. From the mouth, at his eye height, the survey gives 38 m of sight line, and the windrow takes the truck's lower body out of view for the last 25 m of the approach, so a look left from the stopped position would have shown him approximately what a look at 25 km/h showed him. He had done no Bindarra pit driving assessment and had never seen the traffic management plan.
Influence	Pass - very likely (90%)	This is the action that put a light vehicle across a descending haul truck's path, and it is the one place on the map where a different decision in the last ten seconds would have avoided the collision.
Importance	Not tested	(none)

Investigator notes: The object the report must handle most carefully. He is the injured party, he corrected himself on the point least helpful to him, and there was no sign, no line and no assessment to tell him otherwise. Write it as an action, not a failing.

Evidence attached

The main ramp at RL 312 runs approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows, surface dry and well maintained, no corrugation or loose material beyond the normal running surface. The RL 312 dewatering spur joins it from the east on the inside of a left-hand bend, at an included angle of 71 degrees. Marks recovered, measured from the eastern edge of the spur mouth: | Mark | Detail | | --- | --- | | HT-407 left front tyre, braked | Continuous scuff commencing 38.5 m up-ramp of the spur mouth and running 62 m to the truck's resting position | | HT-407 steering | The scuff deviates 1.4 m to the right over its first 20 m, consistent with the 14 degree steering input on the download | | Impact position | 6.2 m into the ramp from the eastern edge of the spur mouth, 3.1 m out from the inner windrow toe | | LV-118 gouge | Continuous gouge from the impact position running 18.4 m down-ramp, changing at 11.2 m from tyre scuff to body panel gouge, which is where the vehicle went over | | LV-118 rest position | On its left side, against the inner windrow, 18.4 m down-ramp of the impact position, facing down-ramp | | Debris | Right-side glass and mirror fragments from 4.8 m to 14 m down-ramp of the impact position | LV-118 damage is concentrated on the right side from the B pillar forward. The roll-over protection structure is deformed inward at the right rear mounting but was not breached. The driver's seatbelt was found fastened and the webbing showed loading marks. The 4.5 m amber beacon whip was sheared at its base and recovered 9 m from the vehicle; its lamp was tested after recovery and worked. HT-407 damage is confined to the front left bumper, the left mudguard and the lower two steps of the access ladder. There is no damage above the bumper line.

Source: INV-2026-0332-SCENE, scene examined 12/08/2026 before recovery

The survey department picked up the intersection, the crest and the eye positions of both vehicles. The spur meets the ramp 34 m beyond the high point of a crest in the ramp. An operator descending the ramp, seated at an eye height of 5.80 m above the running surface, first

has line of sight to the spur mouth at 41 m. A driver at the spur mouth, at an eye height of 1.45 m, first has line of sight to a descending truck at 38 m; the inner windrow, 1.6 m high, stands between the spur and the ramp for the last 25 m of the spur's approach and takes the truck's lower body out of view before that. The Bindarra traffic management plan, clause 6.2, requires that at every intersection the available sight distance shall be not less than the stopping distance of the largest vehicle using the road at the posted speed, and in no case less than 120 m. The stopping distance of a 220 t class truck travelling empty at the posted 40 km/h on a 9 per cent descending grade, calculated with the manufacturer's data and a 1.5 second response, is 96 m. The available sight distance at this intersection is 41 m. The intersection does not meet clause 6.2 on either limb.

Source: Mine survey pickup SUR-26-1188, 13/08/2026, at the investigation's request

Fourteen months at Bindarra, seven years on haul trucks across three sites. Verification of competency on the 220 t class unit issued 06/2025 and current. Annual refresher completed 02/2026. Site general induction 06/2025. The record shows no traffic management refresher of any kind since the general induction, and the general induction module on traffic management is a 23 slide package that names the nine Appendix B intersections and does not mention what to do at an intersection that is not listed. There is no training record anywhere on site, for any person, covering the collision awareness system's suppression geofences. **Light vehicle driver, induction and training record.** Merrivale Survey & Geotechnical training file and the Bindarra gatehouse access card register. Merrivale general mining induction, a two hour online module, completed 24/07/2026. Bindarra site access card issued 09/08/2026 against Merrivale's recognition agreement, following a 20 minute gatehouse briefing covering muster points, speed limits and personal protective equipment. There is no record of the Bindarra site-specific supplement being issued, completed or signed for. There is no record of the Bindarra light vehicle pit driving assessment. Neither record is required by the gatehouse before a card is issued, and neither was asked for. The driver had worked three night shifts at Bindarra: 09/08, 10/08 and 11/08.

Source: Warrigal training file, Merrivale training file and the Bindarra gatehouse access card register

Truck operator braked and steered into the windrow within 1.4 seconds of sighting

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The download records the last steady data point at 02:47:06 (18:47:06 UTC) at 37 km/h, 41 m above the intersection, and service brake pressure rising from 0 to 620 kPa in 0.4 seconds from 02:47:07.4, with the retarder taken to maximum and 14 degrees of right steering input in the same movement. That is a 1.4 second response, inside the 1.5 seconds the plan's own stopping distance calculation allows. Mean deceleration from the brake application to the collision was 1.85 m/s ² against the unit's tested 1.9. Impact speed was 24 km/h; with no braking it would have been 37 km/h. The scene shows the braked scuff deviating 1.4 m right over its first 20 m, into the wall side, which is the only direction that did not take the light vehicle down a 100 m drop with him. He stopped 19 m below the intersection, called the collision, followed the dispatcher's instruction to stay in the cab until the ramp was stopped, then went down and stayed with the driver until the response team arrived.
Influence	Pass - very likely (90%)	The difference between a 37 km/h and a 24 km/h impact on the side of a light vehicle is the difference between the injuries actually sustained and a very different outcome. The steering decision is what kept the vehicle against the inner windrow rather than over the edge.
Importance	Not tested	(none)

Investigator notes: Recorded as a positive event because it is one. The same investigation that finds a 41 m sight line should say that what the operator did inside those four seconds is the reason this is a survivable injury and not a fatality.

Evidence attached

****Haul truck HT-407, descending the main ramp, empty.**** | Time (UTC) | Speed | Position | Note |
 | --- | --- | --- | --- | | 18:45:58 | 41 km/h | RL 368, main ramp | Second cycle after the crib break; tray down, empty | | 18:46:31 | 39 km/h | RL 348 | Retarder on the second position, steady descent | | 18:47:06 | 37 km/h | RL 314, 41 m above the spur intersection | Last data point before the brake application | | 18:47:07.4 | 37 km/h | RL 313 | Service brake pressure rises from 0 to 620 kPa in 0.4 s; retarder to maximum; steering input 14 degrees right | | 18:47:09.2 | 32 km/h | RL 313 | CAS proximity event logged (see below); no alert raised in cab | | 18:47:12.0 | 24 km/h | RL 312, spur intersection | Deceleration spike; collision | | 18:47:14.3 | 0 km/h | 19 m below the intersection | Vehicle stationary, park brake applied | Braking analysis: 1.4 seconds elapsed between the last steady-state data point and the first brake pressure rise, which is the operator's response time on this download. Mean deceleration from the brake application to the collision was 1.85 m/s², against 1.9 m/s² recorded for this unit in the monthly brake performance test on 03/08/2026. The retarder and the service brakes performed as specified throughout. Had no braking been applied the collision speed would have been 37 km/h rather than 24 km/h. Road network: the FMS in-cab map for the Stage 4 pit was last published on 24/07/2026. It does not contain the RL 312 dewatering spur, which was cut on 31/07/2026. The in-cab display therefore showed the main ramp with no intersection at RL 312 on the night of the occurrence. Publishing a revised road network is a survey task raised by a management of change record; no such record exists for the spur. ****Light vehicle LV-118.**** LV-118 is not dispatched and carries no FMS. Its CAS transponder is a passive tag. Its movements are known only from the CAS proximity log, the driver's account and the scene evidence.

Source: Warrigal Mining Services technical services report WMS-TS-2026-114, issued 18/08/2026

The main ramp at RL 312 runs approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows, surface dry and well maintained, no corrugation or loose material beyond the normal running surface. The RL 312 dewatering spur joins it from the east on the inside of a left-hand bend, at an included angle of 71 degrees. Marks recovered, measured from the eastern edge of the spur mouth: | Mark | Detail | | --- | --- | | HT-407 left front tyre, braked | Continuous scuff commencing 38.5 m up-ramp of the spur mouth and running 62 m to the truck's resting position | | HT-407 steering | The scuff deviates 1.4 m to the right over its first 20 m, consistent with the 14 degree steering input on the download | | Impact position | 6.2 m into the ramp from the eastern edge of the spur mouth, 3.1 m out from the inner windrow toe | | LV-118 gouge | Continuous gouge from the impact position running 18.4 m down-ramp, changing at 11.2 m from tyre scuff to body panel gouge, which is where the vehicle went over | | LV-118 rest position | On its left side, against the inner windrow, 18.4 m down-ramp of the impact position, facing down-ramp | | Debris | Right-side glass and mirror fragments from 4.8 m to 14 m down-ramp of the impact position | LV-118 damage is concentrated on the right side from the B pillar forward. The roll-over protection structure is deformed inward at the right rear mounting but was not breached. The driver's seatbelt was found fastened and the webbing showed loading marks. The 4.5 m amber beacon whip was sheared at its base and recovered 9 m from the vehicle; its lamp was tested after recovery and worked. HT-407 damage is confined to the front left bumper, the left mudguard and the lower two steps of the access ladder. There is no damage above the bumper line.

Source: INV-2026-0332-SCENE, scene examined 12/08/2026 before recovery

Occurrence Events

The haul truck's service brakes were degraded and could not achieve the required stopping performance

Test	Result	Supporting text
Existence	Fail - very unlikely (5%)	Hypothesis tested because a 220 tonne truck that does not stop in the distance available raises the brakes first. It fails. The monthly brake performance test on 03/08/2026, eight days before, gave a mean deceleration of 1.9 m/s ² in the empty condition on a 9 per cent grade. The test repeated on 14/08/2026 before the unit returned to service gave 1.92. The download of the occurrence itself gives 1.85 over the 4.6 seconds of braking, which sits between the two tests and inside normal scatter. Service brake pressure rose from 0 to 620 kPa in 0.4 seconds, the retarder was at maximum throughout, and no brake, retarder or park brake defect was raised against the unit in the twelve months before the occurrence. The pre-start check sheet for 11/08/2026 is complete and records no defects. The braking system performed to specification before, during and after the occurrence.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: The first hypothesis anyone offers when a haul truck does not stop, and the first one the records kill. Kept on the map because the report should show it was tested, not assumed away.

Evidence attached

****Haul truck HT-407, descending the main ramp, empty.**** | Time (UTC) | Speed | Position | Note |
 | --- | --- | --- | --- | | 18:45:58 | 41 km/h | RL 368, main ramp | Second cycle after the crib break; tray down, empty | | 18:46:31 | 39 km/h | RL 348 | Retarder on the second position, steady descent | | 18:47:06 | 37 km/h | RL 314, 41 m above the spur intersection | Last data point before the brake application | | 18:47:07.4 | 37 km/h | RL 313 | Service brake pressure rises from 0 to 620 kPa in 0.4 s; retarder to maximum; steering input 14 degrees right | | 18:47:09.2 | 32 km/h | RL 313 | CAS proximity event logged (see below); no alert raised in cab | | 18:47:12.0 | 24 km/h | RL 312, spur intersection | Deceleration spike; collision | | 18:47:14.3 | 0 km/h | 19 m below the intersection | Vehicle stationary, park brake applied | Braking analysis: 1.4 seconds elapsed between the last steady-state data point and the first brake pressure rise, which is the operator's response time on this download. Mean deceleration from the brake application to the collision was 1.85 m/s², against 1.9 m/s² recorded for this unit in the monthly brake performance test on 03/08/2026. The retarder and the service brakes performed as specified throughout. Had no braking been applied the collision speed would have been 37 km/h rather than 24 km/h. Road network: the FMS in-cab map for the Stage 4 pit was last published on 24/07/2026. It does not contain the RL 312 dewatering spur, which was cut on 31/07/2026. The in-cab display therefore showed the main ramp with no intersection at RL 312 on the night of the occurrence. Publishing a revised road network is a survey task raised by a management of change record; no such record exists for the spur. ****Light vehicle LV-118.**** LV-118 is not dispatched and carries no FMS. Its CAS transponder is a passive tag. Its movements are known only from the CAS proximity log, the driver's account and the scene evidence.

Source: Warrigal Mining Services technical services report WMS-TS-2026-114, issued 18/08/2026
 Warrigal Mining Services asset register and maintenance history, obtained 13/08/2026. 220 t class rigid haul truck, commissioned 03/2021, 31,480 engine hours. Last 500 hour service 28/07/2026,

no defects carried. The monthly brake performance test required by the contractor's standard was carried out on 03/08/2026 and passed, with a measured mean deceleration of 1.9 m/s² in the empty condition on a 9 per cent grade. The pre-start check sheet for 11/08/2026 is complete and records no defects. A brake performance test carried out on 14/08/2026 after the collision, before the unit returned to service, gave 1.92 m/s². The retarder, the service brakes and the park brake all met specification before and after the occurrence. No defect was raised against the lighting, the mirrors, the cameras or the cab glass in the twelve months before the occurrence. **Light vehicle LV-118 record.** Corella Resources light vehicle register. 4WD utility, in service 02/2024, 61,200 km. Last periodic service 17/07/2026, 4,100 km before the occurrence. Fitted with a certified roll-over protection structure, a 4.5 m amber beacon whip, reflective marking to the site standard, and a UHF radio. The light vehicle inspection sheet for the week of 09/08/2026 records all items serviceable. The beacon lamp was tested after recovery and worked; the whip was sheared at its base in the roll-over.

Source: Warrigal asset register and Corella light vehicle register, obtained 13/08/2026

Both vehicles arrived at the intersection with neither operator aware of the other

Test	Result	Supporting text
Existence	Pass - extremely likely (95%)	HT-407 was descending the ramp empty at 37 km/h with no intersection shown on its in-cab road network, no warning sign on the approach, no call on the channel and no alert from a collision awareness system that had logged the light vehicle at 21.4 m. LV-118 was leaving the spur at about 25 km/h with 38 m of sight line, no give-way control, no call and no knowledge of the truck. The first sighting available to either of them was at 41 m and about four seconds, which is what the survey predicts for the geometry. Neither the radio log, the download, the CAS log nor any of the three interviews contains a single indication that either operator knew the other was there before that moment. The night shift supervisor did not know a light vehicle was in the pit at all, and there is no permit, call-in or record by which he could have.
Influence	Pass - extremely likely (95%)	Every organisational and control finding on this map converges on this one condition: two vehicles in the same place, at night, each invisible to the other until four seconds remained.
Importance	Not tested	(none)

Investigator notes: The convergence point. Every chain on this map arrives here, which is why the report's analysis should treat it as one mechanism rather than five separate findings.

Evidence attached

****Haul truck HT-407, descending the main ramp, empty.**** | Time (UTC) | Speed | Position | Note |
 | --- | --- | --- | --- | | 18:45:58 | 41 km/h | RL 368, main ramp | Second cycle after the crib break; tray down, empty | | 18:46:31 | 39 km/h | RL 348 | Retarder on the second position, steady descent | | 18:47:06 | 37 km/h | RL 314, 41 m above the spur intersection | Last data point before the brake application | | 18:47:07.4 | 37 km/h | RL 313 | Service brake pressure rises from 0 to 620 kPa in 0.4 s; retarder to maximum; steering input 14 degrees right | | 18:47:09.2 | 32 km/h | RL 313 | CAS proximity event logged (see below); no alert raised in cab | | 18:47:12.0 | 24 km/h | RL 312, spur intersection | Deceleration spike; collision | | 18:47:14.3 | 0 km/h | 19 m below the intersection | Vehicle stationary, park brake applied | Braking analysis: 1.4 seconds elapsed between the last steady-state data point and the first brake pressure rise, which is the operator's response time on this download. Mean deceleration from the brake application to the collision was 1.85 m/s², against 1.9 m/s² recorded for this unit in the monthly brake performance test on 03/08/2026. The retarder and the service brakes performed as specified throughout. Had no braking been applied the collision speed would have been 37 km/h rather than 24 km/h. Road network: the FMS in-cab map for the Stage 4 pit was last published on 24/07/2026. It does not contain the RL 312 dewatering spur, which was cut on 31/07/2026. The in-cab display therefore showed the main ramp with no intersection at RL 312 on the night of the occurrence. Publishing a revised road network is a survey task raised by a management of change record; no such record exists for the spur. ****Light vehicle LV-118.**** LV-118 is not dispatched and carries no FMS. Its CAS transponder is a passive tag. Its movements are known only from the CAS proximity log, the driver's account and the scene evidence.

Source: Warrigal Mining Services technical services report WMS-TS-2026-114, issued 18/08/2026

****Configuration as found on HT-407.**** System operating at Level 7 (operator advisory). Level 9 (automated intervention) is available on this fleet and is not enabled at Bindarra. Advisory alerts are raised on a vehicle tag entering the 60 m detection ring at a closing rate above 10 km/h, except inside a suppression geofence. ****Suppression geofence "S4-LOAD".**** Drawn 14/03/2025 around the Stage 4 loading face to stop nuisance alerts between the excavator, the trucks under it

and the service vehicles working in the same area, where the alert rate made the system unusable. Extended northward on 09/06/2026 when the loading face advanced. The extended geofence boundary lies 180 m up-ramp of the RL 312 spur intersection, so the intersection sits inside the suppressed area. The extension was made by the contractor's technical services group as a system configuration task. It is not recorded in the traffic management plan and was not reviewed against it. **Event log, 11/08/2026.**` 18:47:09.200 PROXIMITY HT-407 tag LV-118 range 21.4 m closing 13 km/h zone S4-LOAD alert SUPPRESSED (geofence) 18:47:12.000 IMPACT HT-407 longitudinal 1.9 g logged by accelerometer 18:47:12.100 PROXIMITY HT-407 tag LV-118 range 0.0 m zone S4-LOAD alert SUPPRESSED (geofence)` The proximity event at 18:47:09.2 was logged 2.8 seconds before the collision, 1.8 seconds after the operator had already begun braking. Outside the geofence the same event would have raised an advisory tone and a screen indication in the cab.

Source: CAS configuration and event log, HT-407, extracted 12/08/2026, WMS-TS-2026-114

The survey department picked up the intersection, the crest and the eye positions of both vehicles. The spur meets the ramp 34 m beyond the high point of a crest in the ramp. An operator descending the ramp, seated at an eye height of 5.80 m above the running surface, first has line of sight to the spur mouth at 41 m. A driver at the spur mouth, at an eye height of 1.45 m, first has line of sight to a descending truck at 38 m; the inner windrow, 1.6 m high, stands between the spur and the ramp for the last 25 m of the spur's approach and takes the truck's lower body out of view before that. The Bindarra traffic management plan, clause 6.2, requires that at every intersection the available sight distance shall be not less than the stopping distance of the largest vehicle using the road at the posted speed, and in no case less than 120 m. The stopping distance of a 220 t class truck travelling empty at the posted 40 km/h on a 9 per cent descending grade, calculated with the manufacturer's data and a 1.5 second response, is 96 m. The available sight distance at this intersection is 41 m. The intersection does not meet clause 6.2 on either limb.

Source: Mine survey pickup SUR-26-1188, 13/08/2026, at the investigation's request

The haul truck's front left struck the right side of the light vehicle at 24 km/h

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Impact at 02:47:12.0 (18:47:12.0 UTC) at 24 km/h, 4.6 seconds after the brake application and 6.2 m into the ramp from the eastern edge of the spur mouth, 3.1 m out from the inner windrow toe. The truck's accelerometer logged 1.9 g longitudinal. Damage to LV-118 is concentrated on the right side from the B pillar forward; damage to HT-407 is confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with nothing above the bumper line. Right side glass and mirror fragments lie from 4.8 m to 14 m down-ramp of the impact position.
Influence	Pass - virtually certain (100%)	A 220 tonne vehicle striking the side of a 3 tonne one leaves the outcome entirely to the light vehicle's structure, the restraint and the speed, and two of those three were already settled before the sighting.
Importance	Not tested	(none)

Investigator notes: The link to make sure the report draws: the truck struck the side of the vehicle, not the front, which is what the intersection geometry makes inevitable and what the roll-over protection then had to deal with.

Evidence attached

****Haul truck HT-407, descending the main ramp, empty.**** | Time (UTC) | Speed | Position | Note |
 | --- | --- | --- | --- | | 18:45:58 | 41 km/h | RL 368, main ramp | Second cycle after the crib break; tray down, empty | | 18:46:31 | 39 km/h | RL 348 | Retarder on the second position, steady descent | | 18:47:06 | 37 km/h | RL 314, 41 m above the spur intersection | Last data point before the brake application | | 18:47:07.4 | 37 km/h | RL 313 | Service brake pressure rises from 0 to 620 kPa in 0.4 s; retarder to maximum; steering input 14 degrees right | | 18:47:09.2 | 32 km/h | RL 313 | CAS proximity event logged (see below); no alert raised in cab | | 18:47:12.0 | 24 km/h | RL 312, spur intersection | Deceleration spike; collision | | 18:47:14.3 | 0 km/h | 19 m below the intersection | Vehicle stationary, park brake applied | Braking analysis: 1.4 seconds elapsed between the last steady-state data point and the first brake pressure rise, which is the operator's response time on this download. Mean deceleration from the brake application to the collision was 1.85 m/s², against 1.9 m/s² recorded for this unit in the monthly brake performance test on 03/08/2026. The retarder and the service brakes performed as specified throughout. Had no braking been applied the collision speed would have been 37 km/h rather than 24 km/h. Road network: the FMS in-cab map for the Stage 4 pit was last published on 24/07/2026. It does not contain the RL 312 dewatering spur, which was cut on 31/07/2026. The in-cab display therefore showed the main ramp with no intersection at RL 312 on the night of the occurrence. Publishing a revised road network is a survey task raised by a management of change record; no such record exists for the spur. ****Light vehicle LV-118.**** LV-118 is not dispatched and carries no FMS. Its CAS transponder is a passive tag. Its movements are known only from the CAS proximity log, the driver's account and the scene evidence.

Source: Warrigal Mining Services technical services report WMS-TS-2026-114, issued 18/08/2026

The main ramp at RL 312 runs approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows, surface dry and well maintained, no corrugation or loose material beyond the normal running surface. The RL 312 dewatering spur joins it from the east on the inside of a left-hand bend, at an included angle of 71 degrees. Marks recovered, measured from the eastern edge of the spur mouth: | Mark | Detail | | --- | --- | | HT-407 left front tyre, braked | Continuous scuff commencing 38.5 m up-ramp of the spur mouth and running 62 m to the truck's resting position | | HT-407 steering | The scuff deviates 1.4 m to the right over its first 20

m, consistent with the 14 degree steering input on the download | | Impact position | 6.2 m into the ramp from the eastern edge of the spur mouth, 3.1 m out from the inner windrow toe | | LV-118 gouge | Continuous gouge from the impact position running 18.4 m down-ramp, changing at 11.2 m from tyre scuff to body panel gouge, which is where the vehicle went over | | LV-118 rest position | On its left side, against the inner windrow, 18.4 m down-ramp of the impact position, facing down-ramp | | Debris | Right-side glass and mirror fragments from 4.8 m to 14 m down-ramp of the impact position | LV-118 damage is concentrated on the right side from the B pillar forward. The roll-over protection structure is deformed inward at the right rear mounting but was not breached. The driver's seatbelt was found fastened and the webbing showed loading marks. The 4.5 m amber beacon whip was sheared at its base and recovered 9 m from the vehicle; its lamp was tested after recovery and worked. HT-407 damage is confined to the front left bumper, the left mudguard and the lower two steps of the access ladder. There is no damage above the bumper line.

Source: INV-2026-0332-SCENE, scene examined 12/08/2026 before recovery

The light vehicle was pushed 18 m down the ramp and rolled onto its left side

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	A continuous gouge runs 18.4 m down-ramp from the impact position, changing at 11.2 m from tyre scuff to body panel gouge, which is where the vehicle went over. LV-118 came to rest on its left side against the inner windrow, facing down-ramp, 18.4 m below the impact. The roll-over protection structure is deformed inward at the right rear mounting and was not breached. The driver's seatbelt was found fastened with loading marks on the webbing. The occupant sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout. The door on the resting side was against the ground, so extrication was through the rear of the vehicle.
Influence	Pass - virtually certain (100%)	The injuries are the ones a side impact and a roll-over produce when the structure holds and the restraint is worn. The same sequence with either of those absent is a different report.
Importance	Not tested	(none)

Investigator notes: The mechanism of injury, and the place to say what worked: the restraint was fastened and loaded, and the roll-over structure deformed without being breached. Both belong in the report.

Evidence attached

HT-407 was descending the main ramp empty, returning to the Stage 4 loading face at the end of the second dump cycle after the crib break. LV-118 was travelling outbound on the RL 312 dewatering spur, a 240 m access road cut under works order WO-26-4471 to reach the new dewatering bore, completed on 31/07/2026 and opened to light vehicles on 02/08/2026. It was returning to the ramp after the grade control technician had completed a pickup at the bore collar. The spur meets the main ramp at a crest. Neither operator saw the other until the two vehicles were about 41 m apart. The haul truck operator applied retard and service braking and steered right, into the windrow. The truck's front left struck the right side of LV-118, pushed it approximately 18 m down the ramp and rolled it onto its left side against the inner windrow. No positive radio call was made by either operator before entering the intersection. The RL 312 spur intersection does not appear in the Bindarra traffic management plan and is not one of the nine intersections at which the plan requires a positive call. The driver of LV-118 was conscious throughout and was extricated by mine emergency response through the rear of the vehicle at 03:14 local, 27 minutes after the collision. ## Immediate actions taken - Stage 4 pit stood down. Main ramp closed below RL 340 and the RL 312 dewatering spur barricaded and closed to all traffic. - Both vehicles isolated and left in position for the investigation. Scene photographed and surveyed before recovery. - Haul truck operator and the shift supervisor stood down for the remainder of the shift. Both tested for drugs and alcohol; both results negative. - Fleet management system and collision awareness system data downloaded from HT-407 by the contractor's technical services group at 09:30 local and provided to the investigation unopened. - Site-wide direction issued by the Mining Manager at 05:10 local: a positive call on the pit channel is required before entering any intersection in the Stage 4 pit, listed in the plan or not, until further notice.

Source: Notification CRL-OCC-2026-0332, raised 12/08/2026 03:26 local (11/08/2026 19:26 UTC)

The main ramp at RL 312 runs approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows, surface dry and well maintained, no corrugation or loose material beyond the normal running surface. The RL 312 dewatering spur joins it from the east on the

inside of a left-hand bend, at an included angle of 71 degrees. Marks recovered, measured from the eastern edge of the spur mouth: | Mark | Detail | | --- | --- | | HT-407 left front tyre, braked | Continuous scuff commencing 38.5 m up-ramp of the spur mouth and running 62 m to the truck's resting position | | HT-407 steering | The scuff deviates 1.4 m to the right over its first 20 m, consistent with the 14 degree steering input on the download | | Impact position | 6.2 m into the ramp from the eastern edge of the spur mouth, 3.1 m out from the inner windrow toe | | LV-118 gouge | Continuous gouge from the impact position running 18.4 m down-ramp, changing at 11.2 m from tyre scuff to body panel gouge, which is where the vehicle went over | | LV-118 rest position | On its left side, against the inner windrow, 18.4 m down-ramp of the impact position, facing down-ramp | | Debris | Right-side glass and mirror fragments from 4.8 m to 14 m down-ramp of the impact position | LV-118 damage is concentrated on the right side from the B pillar forward. The roll-over protection structure is deformed inward at the right rear mounting but was not breached. The driver's seatbelt was found fastened and the webbing showed loading marks. The 4.5 m amber beacon whip was sheared at its base and recovered 9 m from the vehicle; its lamp was tested after recovery and worked. HT-407 damage is confined to the front left bumper, the left mudguard and the lower two steps of the access ladder. There is no damage above the bumper line.

Source: INV-2026-0332-SCENE, scene examined 12/08/2026 before recovery

Light vehicle driver seriously injured in a collision with a haul truck on the main ramp

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	At 02:47:12 on 12/08/2026 (18:47:12 UTC on 11/08/2026) a 220 t class rigid haul truck descending the main ramp of the Stage 4 open pit at Bindarra Gold Mine empty at 24 km/h struck the right side of a light vehicle emerging from the RL 312 dewatering spur, pushed it 18.4 m down the ramp and rolled it onto its left side against the inner windrow. The light vehicle carried one occupant, a grade control technician employed by a survey subcontractor, on his third night shift at the mine. He was conscious throughout, was extricated through the rear of the vehicle 27 minutes after the collision, and was flown to the metropolitan trauma centre at 07:40.
Influence	Pass - virtually certain (100%)	A serious injury: a fractured left clavicle, three fractured left ribs and a small pneumothorax, treated at the surface medical facility and evacuated by air. The light vehicle was destroyed; its roll-over protection structure was deformed inward at the right rear mounting and the seatbelt webbing carried loading marks, so two defences were loaded to the point of showing it. The haul truck was damaged at the front left and returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes and the main ramp below RL 340 was closed for 22 hours. The occurrence was notified to the State Mines Inspectorate as a reportable injury.
Importance	Not tested	(none)

Investigator notes: The critical event. Everything else on this map exists to explain it.

Evidence attached

HT-407 was descending the main ramp empty, returning to the Stage 4 loading face at the end of the second dump cycle after the crib break. LV-118 was travelling outbound on the RL 312 dewatering spur, a 240 m access road cut under works order WO-26-4471 to reach the new dewatering bore, completed on 31/07/2026 and opened to light vehicles on 02/08/2026. It was returning to the ramp after the grade control technician had completed a pickup at the bore collar. The spur meets the main ramp at a crest. Neither operator saw the other until the two vehicles were about 41 m apart. The haul truck operator applied retard and service braking and steered right, into the windrow. The truck's front left struck the right side of LV-118, pushed it approximately 18 m down the ramp and rolled it onto its left side against the inner windrow. No positive radio call was made by either operator before entering the intersection. The RL 312 spur intersection does not appear in the Bindarra traffic management plan and is not one of the nine intersections at which the plan requires a positive call. The driver of LV-118 was conscious throughout and was extricated by mine emergency response through the rear of the vehicle at 03:14 local, 27 minutes after the collision. ## Immediate actions taken - Stage 4 pit stood down. Main ramp closed below RL 340 and the RL 312 dewatering spur barricaded and closed to all traffic. - Both vehicles isolated and left in position for the investigation. Scene photographed and surveyed before recovery. - Haul truck operator and the shift supervisor stood down for the remainder of the shift. Both tested for drugs and alcohol; both results negative. - Fleet management system and collision awareness system data downloaded from HT-407 by the contractor's technical services group at 09:30 local and provided to the investigation unopened. - Site-wide direction issued by the Mining Manager at 05:10 local: a positive call on the pit channel is required before entering any intersection in the Stage 4 pit, listed in the plan or not, until further notice.

Source: Notification CRL-OCC-2026-0332, raised 12/08/2026 03:26 local (11/08/2026 19:26 UTC)

The main ramp at RL 312 runs approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows, surface dry and well maintained, no corrugation or loose material beyond the normal running surface. The RL 312 dewatering spur joins it from the east on the inside of a left-hand bend, at an included angle of 71 degrees. Marks recovered, measured from the eastern edge of the spur mouth: | Mark | Detail | | --- | --- | | HT-407 left front tyre, braked | Continuous scuff commencing 38.5 m up-ramp of the spur mouth and running 62 m to the truck's resting position | | HT-407 steering | The scuff deviates 1.4 m to the right over its first 20 m, consistent with the 14 degree steering input on the download | | Impact position | 6.2 m into the ramp from the eastern edge of the spur mouth, 3.1 m out from the inner windrow toe | | LV-118 gouge | Continuous gouge from the impact position running 18.4 m down-ramp, changing at 11.2 m from tyre scuff to body panel gouge, which is where the vehicle went over | | LV-118 rest position | On its left side, against the inner windrow, 18.4 m down-ramp of the impact position, facing down-ramp | | Debris | Right-side glass and mirror fragments from 4.8 m to 14 m down-ramp of the impact position | LV-118 damage is concentrated on the right side from the B pillar forward. The roll-over protection structure is deformed inward at the right rear mounting but was not breached. The driver's seatbelt was found fastened and the webbing showed loading marks. The 4.5 m amber beacon whip was sheared at its base and recovered 9 m from the vehicle; its lamp was tested after recovery and worked. HT-407 damage is confined to the front left bumper, the left mudguard and the lower two steps of the access ladder. There is no damage above the bumper line.

Source: INV-2026-0332-SCENE, scene examined 12/08/2026 before recovery

Bindarra emergency response log and the post-incident review, obtained 26/08/2026. The night shift is rostered with one qualified mines rescue paramedic, who was at the surface medical facility. The second responder, required before any extrication, is called in from the accommodation village 11 km from the gatehouse. Dispatch paged the emergency response team at 02:48 local. The first response vehicle entered the pit at 02:51 and reached the scene at 02:54. The second responder arrived on scene at 03:07 with the hydraulic rescue tool. The driver was extricated through the rear of the vehicle at 03:14, 27 minutes after the collision, and reached the surface medical facility at 03:41. The post-incident review found the extrication was delayed by approximately 9 minutes waiting for the second responder and the tool, and recorded that the delay did not affect the clinical outcome on this occasion because the injuries were stable and the patient was conscious throughout.

Source: Bindarra emergency response log and post-incident review, obtained 26/08/2026